Symmetrized-MGF Gaussian Rigidity: Chi-Square Quadratic Forms and Sharp Reflection Stability

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Abstract

We study Gaussian rigidity through the reflection-symmetrized moment generating function (MGF). For a centered, variance-one random variable X with a fixed exponential moment, write M_X for its MGF and set

$$R_X(s) = \log M_X(s) + \log M_X(-s) - s^2, \qquad \Delta_\rho(X) = \sup_{|s|\le\rho} |R_X(s)|.$$

We prove an exact rigidity theorem for positive-semidefinite quadratic forms of independent coordinates under the one-sided condition $R_{X_i}(s) \ge 0$, and establish a quantitative reflection-stability theorem under the absolute bound $|R_X(s)| \le \Delta$ on a fixed interval. Specifically, sufficiently small $\Delta_\rho(X)$ implies

$$d_{\mathrm{K}}(X, \mathcal{N}(0,1)) \le C\sqrt{\frac{\log\log(1/\Delta_\rho(X))}{\log(1/\Delta_\rho(X))}}.$$

The proof derives a growing zero-free region for a truncated factor from local reflected-MGF control. A Laguerre construction gives the matching lower order, while a lattice example rules out total-variation stability. Exact quadratic-form and sample-variance results explain how this one-dimensional reflection problem arises from spherical averaging.

1 Introduction

The exact chi-square law of the sample variance is one of the classical signatures of Gaussian samples. The unrestricted converse for a fixed sample size $n \ge 3$ remains a longstanding characterization problem. Existing results impose additional structure, including symmetry or infinite divisibility; see [2, 6, 8, 10–12]. We do not claim to resolve that unrestricted problem. Instead, we isolate the analytic condition that is naturally exposed by spherical averaging and develop its one-dimensional rigidity theory.

The route from the classical question to the present problem is

$$\begin{aligned} \text{sample variance} &\implies \text{chi-square quadratic form} \\ &\implies \text{spherical Laplace averaging} \\ &\implies \text{symmetrized MGF reflection.} \end{aligned}$$

For a centered variance-one random variable X, let $M_X(s) = \mathbb{E}e^{sX}$ whenever finite and define

$$R_X(s) := \log M_X(s) + \log M_X(-s) - s^2, \qquad \Delta_\rho(X) := \sup_{|s|\le\rho} |R_X(s)|. \tag{1}$$

The exact quadratic-form theorem uses the one-sided dominance $R_X(s) \geq 0$, whereas the quantitative reflection theorem only uses the absolute bound $|R_X(s)| \leq \Delta$. We also assume the fixed exponential envelope

$$\mathbb{E}e^{2\tau|X|} \leq K. \tag{2}$$

The reflection has a precise probabilistic meaning. If X' is an independent copy of X, then

$$M_{X-X'}(s) = M_X(s)M_X(-s),$$

so R_X measures the deviation of the reflected convolution from the Gaussian MGF. On the Fourier side the same operation produces $|\varphi_X|^2$ and erases the phase. The central question is therefore whether positivity, exponential integrability, and local control of this phase-blind observable still permit quantitative recovery of the original factor.

Our main quantitative conclusion is the stability estimate

$$d_{\mathrm{K}}(X, \mathcal{N}(0,1)) \leq C_{\rho,\tau,K}\sqrt{\frac{\log\log(1/\Delta)}{\log(1/\Delta)}} \tag{3}$$

for sufficiently small Δ. A Laguerre family in Section 5 shows that this order cannot be improved uniformly in general. Thus the logarithmic scale in (3) is not an artifact of a single low-order expansion; it is forced by high-order odd perturbations whose first m odd moments vanish.

The paper has two complementary exact and quantitative outputs. First, an exact Jensen argument proves Gaussian rigidity for positive-semidefinite chi-square quadratic forms of independent coordinates under coordinatewise dominance; the classical sample-variance characterization with this additional hypothesis is an immediate corollary. Second, the reflection theorem gives a factor-recovery modulus, and the spherical Laplace discrepancy introduced later transfers sample-variance information into its hypotheses. The proof architecture is deliberately modular: local reflected-MGF control gives high-order moments of $X - X'$, those moments lift to a square-root-scale tail for X, truncation produces a bounded factor, and a growing zero-free disk permits quadraticization of its logarithmic characteristic function before Esseen smoothing.

The condition is related to, but distinct from, the standard assumptions in the characterization literature. Infinite divisibility implies the one-sided symmetrized-MGF dominance through the Lévy–Khintchine formula, while symmetry neither implies nor is implied by the dominance. An asymmetric bounded non-infinitely-divisible example and a Rademacher example are given in Section 2. This separation is important because the reflected convolution $F * \widetilde{F}$ loses precisely the phase information retained by the equal-factor convolution $F * F$ studied in classical Cramér stability; see [1, 13]. Related quadratic-form work under symmetry, including Christoph, Prohorov and Ulyanov [3], provides an essential literature boundary but does not assume the present nonsymmetric local reflected-MGF condition.

The remainder of the paper is organized around this single reflection problem. Section 2 defines the symmetrized-MGF class and states the main results. Section 3 gives the complete exact quadratic-form proof. Section 4 contains the five-step stability argument. Section 5 proves the Laguerre lower mechanism and records the total-variation obstruction. Section 6 and Section 7 then return to the sample-variance application and identify the exact spherical observable that feeds the reflection theory. The discussion closes with the remaining open problems rather than with exploratory proof attempts.

2 Symmetrized-MGF dominance and main results

Definition 2.1 (Reflected defect)**.** *Let X be centered with $\mathbb{E}X^2 = 1$, and suppose M_X is finite on $[-\rho, \rho]$. Define*

$$R_X(s) := \log M_X(s) + \log M_X(-s) - s^2, \qquad \Delta_\rho(X) := \sup_{|s|\le\rho} |R_X(s)|.$$

The symmetrized-MGF dominance condition is the one-sided requirement $R_X(s) \ge 0$ on the same interval. Under dominance, $\Delta_\rho(X) = \sup_{|s|\le\rho} R_X(s)$.

Definition 2.2 (Spherical Laplace discrepancy)**.** *For $r \ge 1$, let V be uniform on S^{r-1} and set*

$$\Psi_r(z) := \mathbb{E}_V e^{zV_1}.$$

For a nonnegative random variable Q with the relevant exponential moments, define

$$\mathfrak{D}_{r,\tau}(Q) := \sup_{|t|\le\tau} \left| e^{-t^2/2}\mathbb{E}\Psi_r(t\sqrt{Q}) - 1 \right|.$$

This is the Laplace observable obtained by averaging the radius-$\sqrt{Q}$ transform over a random direction; it is not a generic probability distance.

Proposition 2.3 (Naturality and separation of the dominance condition)**.** *Let X be centered with variance σ^2 and with an MGF in a neighborhood of the origin. If X is infinitely divisible, then*

$$M_X(s)M_X(-s) \ge e^{\sigma^2 s^2}$$

for every s such that both $M_X(s)$ and $M_X(-s)$ are finite. The converse is false, and symmetry alone does not imply this inequality.

Proof. For an infinitely divisible law with finite variance, the Lévy–Khintchine representation gives, near the origin,

$$\log M_X(s) + \log M_X(-s) - \sigma^2 s^2 = 2\int \left(\cosh(sx) - 1 - \frac{s^2x^2}{2}\right) \nu(dx) \ge 0,$$

because $\cosh u - 1 \ge u^2/2$. For the converse, let X take the values 3 and $-1/3$ with probabilities $1/10$ and $9/10$, respectively. It is centered and has variance one. It is not infinitely divisible: if a bounded non-degenerate law of support diameter d were an n-fold convolution of an i.i.d. root, that root would have support diameter at most d/n, and Popoviciu's inequality would give $\mathrm{Var}(X)/n \le d^2/(4n^2)$, which is impossible for large n. Its fourth cumulant is $46/9$, so analyticity at the origin gives

$$R_X(s) = \frac{23}{54}s^4 + O(s^6).$$

Consequently $R_X(s) \ge 0$ on some nontrivial neighborhood of the origin, and this example is asymmetric. Finally, for a Rademacher variable ε, $R_\varepsilon(s) = 2\log\cosh s - s^2 = -s^4/6 + O(s^6)$, which is negative for all sufficiently small nonzero s. □

Theorem 2.4 (Exact PSD rigidity)**.** *Let $X = (X_1, \dots, X_d)$ have independent coordinates with $\mathbb{E}X_i = 0$ and $\mathrm{Var}(X_i) = \sigma_i^2$. Let $A = A^\mathsf{T} \succeq 0$ have rank $r \ge 1$, and suppose*

$$X^\mathsf{T} AX \sim \chi_r^2.$$

For every active coordinate i with $A_{ii} > 0$, assume that $M_i(s) = \mathbb{E}e^{sX_i}$ is finite in a neighborhood of the origin and

$$M_i(s)M_i(-s) \geq e^{\sigma_i^2 s^2}$$

there. Then every active X_i is Gaussian, $X_i \sim \mathcal{N}(0, \sigma_i^2)$.

Corollary 2.5 (Exact sample-variance characterization under dominance)**.** *Let $n \geq 3$ and let $X_1, \dots, X_n$ be i.i.d., centered, and variance one. If M_X is finite near the origin,*

$$M_X(s)M_X(-s) \geq e^{s^2}$$

there, and

$$\sum_{i=1}^{n}(X_i - \bar{X})^2 \sim \chi^2_{n-1},$$

then $X_i \sim \mathcal{N}(0, 1)$.

Theorem 2.6 (Sharp reflection stability)**.** *Fix $\rho, \tau, K > 0$. There exist constants $C < \infty$ and $\Delta_0 \in (0, e^{-e})$, depending only on (ρ, τ, K), such that the following holds. If*

$$\mathbb{E}X = 0, \qquad \mathbb{E}X^2 = 1, \qquad \mathbb{E}e^{2\tau|X|} \leq K,$$

and

$$|R_X(s)| \leq \Delta, \qquad |s| \leq \rho,$$

for some $0 < \Delta \leq \Delta_0$, then

$$d_{\mathrm{K}}(X, \mathcal{N}(0, 1)) \leq C\sqrt{\frac{\log\log(1/\Delta)}{\log(1/\Delta)}}.$$

Theorem 2.7 (Sample-variance stability)**.** *Fix $n \geq 3$, $\tau, K > 0$, and $0 < \rho' < \tau\sqrt{(n-1)/n}$. There are constants $C < \infty$ and $\varepsilon_0 > 0$, depending only on (n, τ, K, ρ'), such that the following holds. Let $X_1, \dots, X_n$ be i.i.d. copies of a centered variance-one X with $\mathbb{E}e^{2\tau|X|} \leq K$, and assume $R_X(s) \geq 0$ for $|s| \leq \tau$. If*

$$Q_n = \sum_{i=1}^{n}(X_i - \bar{X})^2, \qquad \mathfrak{D}_{n-1,\tau}(Q_n) \leq \varepsilon \leq \varepsilon_0,$$

then

$$d_{\mathrm{K}}(X, \mathcal{N}(0, 1)) \leq C\sqrt{\frac{\log\log(1/\varepsilon)}{\log(1/\varepsilon)}}.$$

Theorem 2.8 (Laguerre lower bound)**.** *There are constants $\rho, \tau, K, c, C > 0$ and centered variance-one random variables W_m such that the same uniform exponential-integrability and local dominance assumptions hold, $\Delta_m := \Delta_\rho(W_m) \downarrow 0$, and*

$$\log\frac{1}{\Delta_m} = 2m\log m + O(m), \qquad d_{\mathrm{K}}(W_m, \mathcal{N}(0, 1)) \geq cm^{-1/2}.$$

Consequently

$$d_{\mathrm{K}}(W_m, \mathcal{N}(0, 1)) \geq c'\sqrt{\frac{\log\log(1/\Delta_m)}{\log(1/\Delta_m)}}$$

for all sufficiently large m.

Proposition 2.9 (Strong-distance obstruction)**.** *There is no uniform total-variation stability theorem under the assumptions of Theorem 2.6: a standardized Poisson lattice family can satisfy a fixed exponential-integrability bound and have reflected defect tending to zero while its total-variation distance from every absolutely continuous Gaussian law equals one.*

3 Exact rigidity for chi-square quadratic forms

We prove Theorem 2.4 and then apply it to the ordinary sample variance. Write $A = B^\top B$, where $B \in \mathbb{R}^{r\times d}$ has full row rank, and let b_i be the ith column of B. If V is uniform on S^{r-1}, then spherical averaging gives

$$\mathbb{E}_V \prod_{i=1}^d M_i\big(t\langle b_i, V\rangle\big) = \mathbb{E}_{X,V} e^{t\langle BX,V\rangle} = \mathbb{E}\Psi_r\big(t\sqrt{X^\top AX}\big). \tag{4}$$

Since $X^\top AX \sim \chi^2_r$, the final quantity in (4) equals

$$\mathbb{E}_{G,V} e^{t\langle G,V\rangle} = e^{t^2/2}, \qquad G \sim \mathcal{N}(0, I_r). \tag{5}$$

All expectations below are legitimate for sufficiently small $|t|$, because the coordinate MGFs exist on a common neighborhood of the origin and the sphere is compact.

Define

$$C_i(s) := \log M_i(s) - \frac{\sigma_i^2 s^2}{2}$$

and

$$H_t(v) := \sum_{i=1}^d C_i\big(t\langle b_i, v\rangle\big) + \frac{t^2}{2}\left(\sum_{i=1}^d \sigma_i^2\langle b_i, v\rangle^2 - 1\right). \tag{6}$$

The quadratic terms cancel inside the exponential, so (5) implies

$$\mathbb{E}_V e^{H_t(V)} = e^{-t^2/2}\mathbb{E}_V \prod_i M_i(t\langle b_i, V\rangle) = 1.$$

Jensen's inequality therefore gives $\mathbb{E}_V H_t(V) \le 0$.

The mean of the quadratic correction in (6) is zero. Indeed,

$$\mathbb{E}_V \sum_i \sigma_i^2 \langle b_i, V\rangle^2 = \frac{1}{r}\sum_i \sigma_i^2\|b_i\|^2 = \frac{1}{r}\mathbb{E}(X^\top AX) = 1,$$

where the last equality uses the mean of χ^2_r. Since V and $-V$ have the same law, coordinatewise dominance now gives

$$\begin{aligned}
2\mathbb{E}_V H_t(V) &= \sum_{i=1}^d \mathbb{E}_V\left[\log M_i(t\langle b_i, V\rangle) + \log M_i(-t\langle b_i, V\rangle) - \sigma_i^2 t^2\langle b_i, V\rangle^2\right] \\
&= \sum_{i=1}^d \mathbb{E}_V R_i\big(t\langle b_i, V\rangle\big) \ge 0,
\end{aligned}$$

where $R_i(s) = \log M_i(s) + \log M_i(-s) - \sigma_i^2 s^2$. Hence $\mathbb{E}_V H_t(V) = 0$, and every summand in the last display has zero expectation.

For an active coordinate, $A_{ii} = \|b_i\|^2 > 0$. If $r \ge 2$, the random variable $\langle b_i, V\rangle$ has a density that is strictly positive on the interior of $[-\|b_i\|, \|b_i\|]$. Continuity of R_i therefore implies $R_i(s) = 0$ on a neighborhood of the origin. If $r = 1$, then $V \in \{-1, 1\}$ and varying t directly gives the same conclusion, because R_i is even. Thus, for every active coordinate,

$$M_i(s)M_i(-s) = e^{\sigma_i^2 s^2}$$

near the origin. If $\sigma_i^2 = 0$, then $X_i = 0$ almost surely. If $\sigma_i^2 > 0$, an independent copy X_i' gives that $X_i - X_i'$ has the MGF of $\mathcal{N}(0, 2\sigma_i^2)$ near zero and hence has that Gaussian law.

The Lévy–Crámer decomposition theorem applied to the independent sum $X_i + (-X_i')$ then implies that X_i is Gaussian.

This proves Theorem 2.4. For Corollary 2.5, take

$$A = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^{\mathsf{T}}.$$

It is a rank-$n-1$ orthogonal projection, every diagonal entry is $(n-1)/n > 0$, and its quadratic form is exactly $\sum_i (X_i - \bar{X})^2$. The corollary follows immediately.

4 Sharp reflection stability

Throughout this section let X' be an independent copy of X, set

$$Y = X - X', \qquad G \sim \mathcal{N}(0, 2),$$

and define

$$F(z) := M_Y(z) - e^{z^2}, \qquad L := \log(1/\Delta).$$

We prove Theorem 2.6 through five lemmas.

4.1 High-order reflected moment transfer

Lemma 4.1 (Local defect to high-order moments)**.** *Fix ρ, τ, K. For every $A > 0$ there is $c_0 = c_0(A, \rho, \tau, K) > 0$ such that, with*

$$m = \left\lfloor c_0 \frac{L}{\log L} \right\rfloor,$$

all sufficiently small Δ satisfy

$$\left| \mathbb{E}Y^k - \mathbb{E}G^k \right| \le e^{-Am\log m}, \qquad 0 \le k \le 2m.$$

In particular,

$$\mathbb{E}|Y|^{2m} \le (Cm)^m.$$

Proof. Choose a fixed $0 < \rho_0 < \min\{\rho, \tau\}$ and a symmetric ellipse whose upper and lower half-ellipses lie in the strip $|\Re z| < 2\tau$. Exponential integrability gives a bound $|F(z)| \le B_0$ on this ellipse, with B_0 depending only on (ρ, τ, K). On the real diameter, the defect hypothesis gives

$$|F(s)| = e^{s^2}|e^{R_X(s)} - 1| \le C_0\Delta, \qquad |s| \le \rho_0,$$

where $F(s) = M_Y(s) - e^{s^2}$ and $Y = X - X'$. To make the propagation step explicit, apply the subharmonic two-constants theorem to $u(z) = \log|F(z)|$ on each half-ellipse. If $\omega(z)$ denotes the harmonic measure of the real diameter, then

$$u(z) \le \omega(z)\log(C_0\Delta) + (1 - \omega(z))\log B_0.$$

The harmonic-measure lower bound on a fixed inner disk gives $\omega(z) \ge \omega_0 > 0$ whenever $|z| \le r_0$, after possibly shrinking r_0. Thus there are constants $r_0 > 0$, $\omega_0 > 0$, and $C_1 < \infty$ such that

$$\sup_{|z|\le r_0} |F(z)| \le C_1\Delta^{\omega_0}.$$

The Cauchy estimate consequently gives

$$|F^{(k)}(0)| \le C_1 k! r_0^{-k} \Delta^{\omega_0}.$$

For $k \le 2m$ and $m = c_0 L/\log L$, Stirling's bound shows that the logarithm of the right-hand side is at most $-\omega_0 L + O(c_0 L)$. Choosing c_0 so that $(A+3)c_0 < \omega_0$ gives $|F^{(k)}(0)| \le e^{-Am\log m}$. Since $F^{(k)}(0) = \mathbb{E}Y^k - \mathbb{E}G^k$, the moment conclusion follows. Finally,

$$\mathbb{E}G^{2m} = \frac{(2m)!}{m!} \le (Cm)^m,$$

and the negligible moment mismatch gives the stated $2m$-th moment bound. □

4.2 Tail lifting and bounded truncation

Lemma 4.2 (Square-root tail lifting)**.** *There are constants $C, c > 0$ with the following property. For every sufficiently large fixed B, if $N = B\sqrt{m}$, then*

$$\mathbb{P}(|X| > N) \le Ce^{-\beta_B m},$$

where $\beta_B \to \infty$ as $B \to \infty$. If X_N denotes the conditional law $X \mid \{|X| \le N\}$, then

$$d_{\mathrm{TV}}(X, X_N) \le Ce^{-\beta_B m}.$$

Moreover

$$|\mathbb{E}X_N| + |\operatorname{Var}(X_N) - 1| \le C(1+N^2)e^{-cN}.$$

Proof. Let a be a median of X. Since $\mathbb{E}X^2 = 1$, $|a| \le \sqrt{2}$. The safe symmetrization estimate is

$$\mathbb{P}(|X-a| > x) \le 4\mathbb{P}(|X - X'| > x).$$

Combine this with Markov's inequality and Lemma 4.1. The conditional-law total-variation estimate is immediate. The mean and variance shifts are controlled separately from the original exponential moment, which gives exponentially small first and second tail moments at radius N. □

4.3 A growing real Gaussian-product window

Lemma 4.3 (Real-axis product approximation)**.** *For B fixed sufficiently large, there is $a_0 = a_0(B, \rho, \tau, K) > 0$ such that, with*

$$R = a_0\sqrt{m}$$

and φ_N the characteristic function of X_N,

$$\sup_{|t|\le 2R} \left|\varphi_N(t)\varphi_N(-t) - e^{-t^2}\right| \le e^{-\gamma_1 m}$$

for some $\gamma_1 > 0$.

Proof. Use the $2m-1$ Taylor polynomial of the characteristic function of $Y = X - X'$ and the real-variable remainder

$$\left|e^{iu} - \sum_{k=0}^{2m-1} \frac{(iu)^k}{k!}\right| \le \frac{|u|^{2m}}{(2m)!}.$$

For clarity, the comparison has three contributions:

$$
\begin{aligned}
|\varphi_Y(t) - e^{-t^2}| \le{}& \underbrace{\frac{|t|^{2m}\mathbb{E}|Y|^{2m}}{(2m)!}}_{Y\text{ remainder}} + \underbrace{\frac{|t|^{2m}\mathbb{E}|G|^{2m}}{(2m)!}}_{\text{Gaussian remainder}} \\
&+ \underbrace{\sum_{k=0}^{2m-1} \frac{|t|^k}{k!} |\mathbb{E}(iY)^k - \mathbb{E}(iG)^k|}_{\text{moment mismatch}},
\end{aligned}
$$

where $G \sim \mathcal{N}(0,2)$. On $|t| \le 2R$, the first two terms are bounded by $(Ca_0^2)^m$ after Stirling, while the third is super-exponentially smaller by Lemma 4.1. Choose a_0 small. Since $\varphi_Y(t) = \varphi_X(t)\varphi_X(-t)$ and the truncation error is bounded by Lemma 4.2, transfer to X_N gives

$$\varphi_N(t)\varphi_N(-t) = \varphi_{X_N - X_N'}(t) = e^{-t^2} + O(e^{-\gamma_1 m}).$$

□

4.4 Complex propagation and a zero-free factor disk

Lemma 4.4 (Growing zero-free disk)**.** *The constants B and a_0 can be chosen so that, for $R = a_0\sqrt{m}$,*

$$\varphi_N(z) \neq 0, \qquad |z| \le R.$$

More precisely, for some $\gamma_2 > a_0^2$,

$$\sup_{|z|\le R} \left| \varphi_N(z)\varphi_N(-z) - e^{-z^2} \right| \le e^{-\gamma_2 m}.$$

Proof. Since $|X_N| \le N$, φ_N is entire and

$$|\varphi_N(z)| \le e^{N|\Im z|}.$$

Hence, for

$$\mathcal{F}_N(z) := \varphi_N(z)\varphi_N(-z) - e^{-z^2},$$

the explicit growth bound

$$\log|\mathcal{F}_N(z)| \le \log 2 + 4NR + 4R^2 \le \log 2 + 4(Ba_0 + a_0^2)m, \qquad |z| \le 2R,$$

holds. On the real diameter, Lemma 4.3 gives $\log|\mathcal{F}_N| \le -\gamma_1 m$. Apply the subharmonic two-constants theorem separately in the upper and lower half-disks of radius $2R$. The harmonic measure of the diameter is uniformly bounded below by a fixed $\omega_0 > 0$ on the concentric half-disks of radius R, so

$$\log|\mathcal{F}_N(z)| \le -\left[\omega_0\gamma_1 - 4(1-\omega_0)(Ba_0 + a_0^2)\right] m + O(1).$$

Choose first B large and then a_0 sufficiently small so that the bracket is a number $\gamma_2 > a_0^2$. Since

$$\inf_{|z|\le R} |e^{-z^2}| \ge e^{-R^2} = e^{-a_0^2 m},$$

relative closeness implies $\varphi_N(z)\varphi_N(-z) \neq 0$, hence $\varphi_N(z) \neq 0$ on the disk. □

4.5 Logarithmic Gaussianization and Esseen smoothing

Lemma 4.5 (Factor Gaussianization)**.** *Under the conclusions of Lemma 4.4,*

$$d_{\mathrm{K}}(X_N, \mathcal{N}(\mu_N, \sigma_N^2)) \le \frac{C}{R},$$

where $\mu_N = \mathbb{E}X_N$ *and* $\sigma_N^2 = \mathrm{Var}(X_N)$*. Consequently*

$$d_{\mathrm{K}}(X, \mathcal{N}(0,1)) \le \frac{C}{\sqrt{m}}.$$

Proof. On $|z| < R$ choose the analytic logarithm

$$\psi_N(z) = \log \varphi_N(z), \qquad \psi_N(0) = 0.$$

The bounded support gives $\Re\psi_N(z) = \log|\varphi_N(z)| \le NR$. Set

$$H(z) := NR - \psi_N(z), \qquad \Re H(z) \ge 0.$$

After scaling $z = Rw$, the Carathéodory coefficient inequality applied to $H(Rw)/(NR)$ gives, for $\psi_N(z) = \sum_{k\ge1} a_k z^k$,

$$|a_k| \le 2NR^{1-k}.$$

Since $a_1 = i\mu_N$ and $a_2 = -\sigma_N^2/2$, for every fixed $0 < \theta < 1$,

$$\left|\psi_N(t) - i\mu_N t + \frac{\sigma_N^2 t^2}{2}\right| \le \sum_{k\ge3} 2NR^{1-k}|t|^k \le \frac{2N|t|^3}{R^2(1-|t|/R)} \le C\frac{|t|^3}{R}, \qquad |t| \le \theta R,$$

because $N/R = B/a_0$ is fixed. By Lemma 4.2, $\mu_N = o(R^{-1})$ and $\sigma_N^2 = 1 + o(R^{-1})$. Taking θ sufficiently small gives Gaussian damping, and hence

$$\left|\varphi_N(t) - e^{i\mu_N t - \sigma_N^2 t^2/2}\right| \le C\frac{|t|^3}{R}e^{-ct^2}.$$

The Esseen inequality in the present normalization is

$$d_{\mathrm{K}}(X_N, \mathcal{N}(\mu_N, \sigma_N^2)) \le \frac{1}{\pi}\int_{-T}^{T}\left|\frac{\varphi_N(t) - e^{i\mu_N t - \sigma_N^2 t^2/2}}{t}\right| dt + \frac{C}{T}.$$

With $T = \theta R$, the integral is $O(R^{-1})$ because its integrand is bounded by $Ct^2 e^{-ct^2}/R$, and the cutoff term is also $O(R^{-1})$. Add the truncation and parameter-shift errors. ☐

Proof of Theorem 2.6. Apply Lemmas 4.1, 4.2, 4.3, 4.4, and 4.5. Since

$$m \asymp \frac{L}{\log L}, \qquad L = \log(1/\Delta),$$

we have

$$\frac{1}{\sqrt{m}} \asymp \sqrt{\frac{\log L}{L}},$$

which is the claimed rate. ☐

5 Optimality and obstruction results

5.1 A Laguerre lower-bound family

We give the lower-bound construction in full because it identifies the scale in Theorem 2.6. The only analytic input needed for the probability construction is the following uniform Laguerre envelope; its normalization and four-region proof are recorded in Appendix B.

Lemma 5.1 (Uniform Laguerre envelope)**.**

$$C_L := \sup_{m\geq 1,\, x\in\mathbb{R}} \left| xe^{-x^2} L_m^{(1/2)}(3x^2/2) \right| < \infty. \tag{7}$$

Proof. The four-region estimate and the conversion from the normalized Laguerre function are given in Appendix B. □

Fix $0 < \epsilon < 1/4$, let ϕ be the standard Gaussian density, and define

$$p_m(x) := xe^{-x^2} L_m^{(1/2)}(3x^2/2), \qquad h_m(x) := \frac{p_m(x)}{C_L}, \qquad f_m(x) := \phi(x)(1 + \epsilon h_m(x)).$$

By (7), f_m is a nonnegative probability density. It is normalized because h_m is odd. The change of variables $r = 3x^2/2$ and Laguerre orthogonality show that

$$\int_{\mathbb{R}} x^{2k+1} h_m(x)\phi(x)\, dx = 0, \qquad 0 \leq k < m.$$

In particular, the corresponding variable U_m is centered. Its variance is one because $x^2 h_m(x)\phi(x)$ is odd. The Laguerre generating identity gives the exact MGF

$$M_{U_m}(s) = e^{s^2/2}(1 + u_m(s)), \qquad u_m(s) = \frac{\epsilon}{C_L} \frac{s}{3\sqrt{3}} e^{-s^2/3} \frac{(-s^2/6)^m}{m!}. \tag{8}$$

Therefore, with

$$a_m := \frac{\epsilon}{3\sqrt{3}C_L 6^m m!},$$

we have $|u_m(s)| \leq a_m |s|^{2m+1}$ on the real axis.

Let $J_m = N_m^+ - N_m^-$, where N_m^+ and N_m^- are independent Poisson variables with mean $\lambda_m/2$, and set

$$\lambda_m := 96 a_m^2 \rho^{4m-2}, \qquad W_m := \frac{U_m + J_m}{\sqrt{1 + \lambda_m}}.$$

Then W_m is centered and variance one. If $q = s/\sqrt{1 + \lambda_m}$, its reflected defect is exactly

$$R_{W_m}(s) = \log(1 - u_m(q)^2) + 2\lambda_m \left(\cosh q - 1 - \frac{q^2}{2} \right). \tag{9}$$

For all sufficiently large m, $|u_m(q)| \leq 1/2$ on $|s| \leq \rho$. Using $-\log(1 - v) \leq 2v$ for $0 \leq v \leq 1/4$ and $\cosh q - 1 - q^2/2 \geq q^4/24$, the choice of λ_m gives

$$R_{W_m}(s) \geq \frac{\lambda_m q^4}{16} \geq 0.$$

The logarithmic term in (9) is nonpositive, so the upper Taylor bound gives

$$R_{W_m}(s) \leq \frac{\lambda_m \cosh(\rho) q^4}{12}.$$

If also $\lambda_m \le 1$, evaluation at $s = \rho$ yields

$$\frac{\lambda_m \rho^4}{64} \le \Delta_m := \Delta_\rho(W_m) \le \frac{\lambda_m \rho^4 \cosh(\rho)}{12}.$$

Thus $\Delta_m \asymp \lambda_m$ and Stirling's formula gives

$$\log \frac{1}{\Delta_m} = 2m \log m + O(m).$$

The fixed exponential envelope is uniform: $f_m \le (1+\epsilon)\phi$, while

$$\mathbb{E} e^{b|J_m|} \le 2e^{\lambda_m(\cosh b - 1)}$$

and $\sup_m \lambda_m < \infty$. Finally, the half-axis discrepancy is explicit. Since

$$\int_0^\infty h_m(x)\phi(x)\,dx = \frac{1}{3C_L\sqrt{2\pi}} \frac{(1/2)_m}{m!},$$

we have $|F_{U_m}(0) - 1/2| \asymp m^{-1/2}$. Moreover,

$$|F_{U_m+J_m}(0) - F_{U_m}(0)| \le \mathbb{P}(J_m \ne 0) \le \lambda_m = o(m^{-1/2}).$$

The normalization in W_m leaves the threshold zero unchanged. Consequently

$$d_{\mathrm{K}}(W_m, \mathcal{N}(0,1)) \ge c m^{-1/2}.$$

Together with $\log(1/\Delta_m) = 2m \log m + O(m)$, this proves Theorem 2.8. It also rules out every uniform power-law modulus $C\Delta^\alpha$, $\alpha > 0$, and rules out the previously proposed $C/\sqrt{\log(1/\Delta)}$ bound.

5.2 Failure of total-variation stability

Let $N_\lambda \sim \mathrm{Poisson}(\lambda)$ and $X_\lambda = (N_\lambda - \lambda)/\sqrt{\lambda}$. The reflected defect satisfies, on every fixed compact s-interval,

$$R_{X_\lambda}(s) = \frac{s^4}{12\lambda} + O_\rho(\lambda^{-2}),$$

and the family has a fixed exponential-integrability bound for each fixed τ. Nevertheless X_λ is lattice-valued, whereas $\mathcal{N}(0,1)$ is absolutely continuous, so

$$d_{\mathrm{TV}}(X_\lambda, \mathcal{N}(0,1)) = 1.$$

This is a qualitative boundary phenomenon and is independent of the Kolmogorov lower bound above.

6 From spherical Laplace discrepancy to reflected defect

We now identify the observable generated by the sample-variance quadratic form. Let $r = n-1$, let $P = I_n - n^{-1}\mathbf{1}\mathbf{1}^\top$, and choose $B \in \mathbb{R}^{r \times n}$ with $B^\top B = P$ and $BB^\top = I_r$. If V is uniform on S^{r-1}, then every column b_i of B has norm

$$q := \|b_i\| = \sqrt{\frac{n-1}{n}},$$

and $\langle b_i, V\rangle$ has the same distribution as qV_1. Put $a = \tau q$ and define

$$D_X(s) := \frac{1}{2}R_X(s) = \frac{1}{2}\log\frac{M_X(s)M_X(-s)}{e^{s^2}}.$$

Under dominance, $D_X \geq 0$ on $[-\tau, \tau]$.

The exact definition of the discrepancy in Section 2 gives, for $|t| \leq \tau$,

$$e^{-t^2/2}\mathbb{E}\Psi_r(t\sqrt{Q_n}) = \mathbb{E}_V \exp\left(\sum_{i=1}^{n}\left[\log M_X(t\langle b_i, V\rangle) - \frac{t^2\langle b_i, V\rangle^2}{2}\right]\right).$$

Jensen's inequality, exchangeability, and the symmetry of V_1 therefore imply

$$n\mathbb{E}D_X(aV_1) \leq \log\bigl(1 + \mathfrak{D}_{n-1,\tau}(Q_n)\bigr). \tag{10}$$

This is the precise point at which the spherical observable enters the one-dimensional reflection theory.

For completeness, we record the interior inversion of (10). The density of V_1 is

$$f_n(v) = \frac{\Gamma((n-1)/2)}{\sqrt{\pi}\,\Gamma((n-2)/2)}(1-v^2)^{(n-4)/2}, \qquad -1 < v < 1.$$

Fix $0 < \rho' < a$ and choose $0 < h < a - \rho'$. On $[-(\rho' + h), \rho' + h]$ the scaled density of aV_1 is bounded below by

$$m_h := \frac{1}{a}\frac{\Gamma((n-1)/2)}{\sqrt{\pi}\,\Gamma((n-2)/2)}\left(1 - \left(\frac{\rho' + h}{a}\right)^2\right)^{\max(n-4,0)/2}.$$

The exponential envelope implies that D_X is Lipschitz there. One may take

$$L_h := \frac{K}{e\tau} + \rho' + h,$$

because $M_X(s) \geq 1$ and $|x|e^{\tau|x|} \leq (e\tau)^{-1}e^{2\tau|x|}$. If $M_* := \sup_{|s|\leq\rho'} D_X(s)$, two disjoint Lipschitz tents centered at the points where the maximum is attained give

$$2m_h I_{L_h,h}(M_*) \leq \frac{1}{n}\log\bigl(1 + \mathfrak{D}_{n-1,\tau}(Q_n)\bigr).$$

Here the tent area is the nonnegative piecewise function

$$I_{L,h}(M) := \begin{cases} M^2/L, & M \leq Lh, \\ 2hM - Lh^2, & M > Lh. \end{cases}$$

Writing $G(y; L, h) = \sqrt{Ly}$ when $y \leq Lh^2$ and $G(y; L, h) = y/(2h) + Lh/2$ otherwise, we obtain

$$\sup_{|s|\leq\rho'} D_X(s) \leq \inf_{0<h<a-\rho'} G\left(\frac{\log(1 + \mathfrak{D}_{n-1,\tau}(Q_n))}{2nm_h}; L_h, h\right). \tag{11}$$

In particular, for fixed n, τ, K, ρ', the right-hand side is $O(\sqrt{\mathfrak{D}_{n-1,\tau}(Q_n)})$ as the discrepancy tends to zero. The strict interior restriction is essential when $n \geq 5$, because the spherical density vanishes at the endpoints.

7 Sample variance as an application

Let $X_1, \dots, X_n$ be i.i.d. copies of a centered variance-one random variable X, and let

$$Q_n = \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

The spherical Laplace discrepancy $\mathfrak{D}_{n-1,\tau}(Q_n)$ is the natural input because Q_n is the squared radius of the projection of $(X_1, \dots, X_n)$ onto the hyperplane orthogonal to $(1, \dots, 1)$. If $\mathfrak{D}_{n-1,\tau}(Q_n) \le \varepsilon$, the pointwise inversion in (11) gives, for every fixed $0 < \rho' < \tau\sqrt{(n-1)/n}$,

$$\sup_{|s| \le \rho'} |R_X(s)| = 2 \sup_{|s| \le \rho'} D_X(s) \le C\sqrt{\varepsilon}.$$

Theorem 2.6 then yields

$$d_{\mathrm{K}}(X, \mathcal{N}(0,1)) \le C\sqrt{\frac{\log\log(1/\varepsilon)}{\log(1/\varepsilon)}}$$

for sufficiently small ε, where fixed multiplicative constants inside the logarithms are absorbed into C. This proves Theorem 2.7.

The exact case is even shorter. If $Q_n \sim \chi^2_{n-1}$ and the local dominance holds, Corollary 2.5 gives $X_i \sim \mathcal{N}(0,1)$. Thus the discrepancy-to-defect argument is a quantitative extension of the exact spherical-Jensen mechanism, rather than a separate sample-variance characterization.

8 Discussion and open directions

The paper separates three logically different statements. The exact quadratic-form theorem is a Jensen-equality rigidity result under a one-sided coordinatewise order. The sharp reflection theorem is a one-dimensional factor-recovery statement and does not require the order sign. The sample-variance result is the interface between them: spherical averaging creates the reflected MGF observable, and the interior tent inversion converts its integrated control into a pointwise defect. Keeping these layers separate makes clear both the scope and the novelty boundary of the work.

The unrestricted fixed-n sample-variance characterization remains outside the scope of the paper. Removing local symmetrized-MGF dominance would require a mechanism that recovers the odd phase information from the quadratic statistic alone. The present results show that the additional condition is natural and useful, but they do not turn it into a consequence of exact sample-variance information.

Three questions remain open. First, can the fixed exponential-integrability assumption be weakened to a sharp tail or moment hypothesis without changing the reflection modulus? Second, under which additional uniform-integrability hypothesis can a standard distributional distance replace the spherical Laplace discrepancy while retaining the same interior pointwise transfer? Third, can the factor-recovery and optimality theory be extended from the i.i.d. sample-variance projection to general positive-semidefinite quadratic forms with non-identically distributed coordinates?

A Parameter selection for the sharp upper bound

This appendix records a non-circular order of choices for the five-step proof. Fix (ρ, τ, K) and an interior analytic radius. Let $\omega_0 > 0$ be the lower bound for the harmonic measure of the real diameter in the smaller half-disk used in the propagation step, and let C_G be the corresponding complex-growth constant. Choose $A > 8$ and then choose $c_0 > 0$ so small that

$$(A+3)c_0 < \omega_0.$$

With $m = \lfloor c_0 L/\log L \rfloor$, this makes the fixed-disk Cauchy estimate dominate all factorial losses in the moment-transfer lemma.

Next choose the truncation factor B sufficiently large that the square-root tail estimate has exponent $\beta_B > 4$. After B is fixed, choose $a_0 > 0$ so small that the real Taylor remainder obeys $C_T a_0^2 < e^{-4}$ and, with $\gamma_1 = -\log(C_T a_0^2)$,

$$C_G(Ba_0 + a_0^2) < \frac{\omega_0 \gamma_1}{2}, \qquad \gamma_2 := \omega_0\gamma_1 - C_G(Ba_0 + a_0^2) > a_0^2.$$

Such an a_0 exists because the left-hand growth term tends to zero while γ_1 tends to infinity as $a_0 \downarrow 0$. Finally choose the fixed Esseen fraction $0 < \theta < 1/4$ so that the cubic remainder in the logarithmic expansion is at most one quarter of the Gaussian quadratic damping on $|t| \le \theta R$. All constants then depend only on the fixed model parameters and not on Δ or m.

B Laguerre estimates and construction details

For reference, Imekraz–Robert–Thomann normalize the Laguerre function by

$$\mathcal{L}_m^{(\alpha)}(r) = a_{m,\alpha} L_m^{(\alpha)}(r) e^{-r/2} r^{\alpha/2}, \qquad a_{m,\alpha} = \sqrt{\frac{m!}{\Gamma(m+\alpha+1)}}, \qquad \nu = 4m + 2\alpha + 2.$$

For $\alpha = 1/2$, Proposition 3.2 of [9] gives the four estimates

$$\begin{aligned} |\mathcal{L}_m^{(1/2)}(r)| &\le C(r\nu)^{1/4} \quad (0 \le r \le \nu^{-1}),\\ |\mathcal{L}_m^{(1/2)}(r)| &\le C(r\nu)^{-1/4} \quad (\nu^{-1} \le r \le \nu/2),\\ |\mathcal{L}_m^{(1/2)}(r)| &\le C\nu^{-1/4}\big(\nu^{1/3} + |\nu - r|\big)^{-1/4} \quad (\nu/2 \le r \le 3\nu/2), \end{aligned}$$

and an exponential estimate for $r \ge 3\nu/2$. Since

$$\sqrt{r}\, e^{-r/2} |L_m^{(1/2)}(r)| = \sqrt{\frac{\Gamma(m+3/2)}{m!}}\, r^{1/4} |\mathcal{L}_m^{(1/2)}(r)|$$

and the gamma ratio is comparable to $\nu^{1/4}$, the first two regions give a uniform bound, the turning region gives $O(\nu^{1/6})$, and the outer region is exponentially decaying. Multiplication by $e^{-r/6}$ therefore yields

$$\sup_{m\ge 1,\, r \ge 0} \sqrt{r}\, e^{-2r/3} |L_m^{(1/2)}(r)| < \infty.$$

The remaining identities used in Section 5 are elementary consequences of the Laguerre generating function and orthogonality. In particular, the perturbation is odd, the first m odd moments vanish, and the exact MGF is the expression in (8). The Skellam compensation is chosen after the MGF calculation so that the positive fourth-order hyperbolic remainder dominates the negative term $\log(1 - u_m(q)^2)$ pointwise. This order of construction is what makes the one-sided dominance check independent of an asymptotic sign guess.

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